

COM3110/4115: Text Processing

Information Retrieval: Retrieval Models

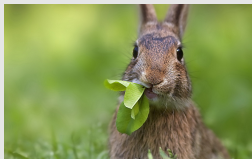
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- Definition of the information retrieval problem
- Approaches to document indexing
 - ◊ manual approaches
 - ◊ automatic approaches
- **Automated retrieval models**
 - ◊ **boolean model**
 - ◊ **ranked retrieval methods (e.g. vector space model)**
- Term manipulation:
 - ◊ stemming, stopwords, term weighting
- Evaluation
- Advanced Topics in IR

Document Representation: Bag-of-Words Approach

- Traditional approach to representing documents (and queries) in IR:
 - ◇ record what words (terms) are present
 - ◇ usually, plus count of term in each document
 - ◇ can be easily stored in an **inverted index** for efficient access
- Ignores relations between words
 - ◇ i.e. of order, proximity, etc.
 - ◇ e.g. rabbit eating = eating rabbit



- Such representations known as **bag of words** approaches
 - ◇ c.f. mathematical structure “bag” (aka “multiset”)
— like a set (i.e. unordered), but records a count for each element

Approaches to Document-Query Matching

- **Boolean retrieval:**

- ◇ match between query and document a **binary decision**: is document relevant to query or not?
- ◇ presence of search term in document necessary and sufficient for match

- **Ranked retrieval:**

- ◇ match between query and document a **matter of degree**
- ◇ frequency of document terms taken into account
- ◇ not all search terms necessarily present in document
- ◇ Incarnations:
 - **The vector space model (SMART, Salton et al, 1971)**
 - The probabilistic model (OKAPI, Robertson/Spärck Jones, 1976)
 - Web search engines

The Boolean model

- Approach: construct *complex search commands*, by
 - ◊ combining *basic* search terms (keywords)
 - ◊ using *boolean operators*
 - NB: this is NOT why approach is called boolean
- *Boolean Operators*:
 - ◊ AND, OR, NOT, BUT, XOR (*exclusive* OR)
- E.g.:

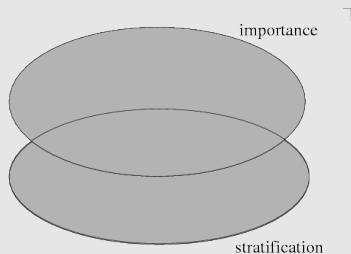
Monte-Carlo AND (importance OR stratification) BUT gambling
- Boolean query provides a simple logical basis for deciding whether any document should be returned, based on:
 - ◊ whether basic terms of query do/do not appear in the document
 - ◊ the meaning of the logical operators

The Boolean model: set-theoretic interpretation

- Boolean operators have a **set-theoretic interpretation** for **efficient** retrieval
- Overall document collection forms **maximal document set**
- let $d(E)$ denote the document set for expression E
 - ◇ E either a basic term or boolean expression
- Boolean operators map to set-theoretic operations:
 - ◇ AND $\mapsto \cap$ (intersection): $d(E_1 \text{ AND } E_2) = d(E_1) \cap d(E_2)$
 - ◇ OR $\mapsto \cup$ (union): $d(E_1 \text{ OR } E_2) = d(E_1) \cup d(E_2)$
 - ◇ NOT $\mapsto ^c$ (complement): $d(\text{NOT } E) = d(E)^c$
 - ◇ BUT $\mapsto -$ (difference): $d(E_1 \text{ BUT } E_2) = d(E_1) - d(E_2)$

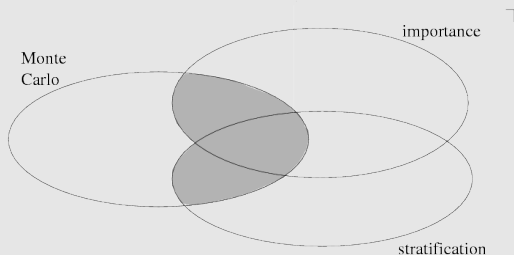
The Boolean model: set-theoretic interpretation (contd)

E.g. Monte-Carlo AND (importance OR stratification) BUT gambling



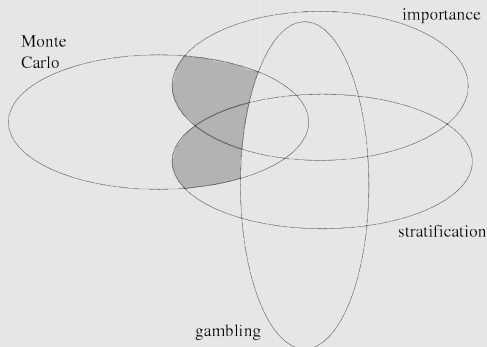
The Boolean model: set-theoretic interpretation (contd)

E.g. **Monte-Carlo AND (importance OR stratification) BUT gambling**



The Boolean model: set-theoretic interpretation (contd)

E.g. **Monte-Carlo AND (importance OR stratification) BUT gambling**



Boolean Queries: Complexity

- Question: **Magnetic resonance imaging, magnetic resonance arthrography and ultrasonography for assessing rotator cuff tears in people with shoulder pain for whom surgery is being considered**
- Query: ((Ultrasonography [mh] OR ultrasound [tw] OR ultrasonograph* [tw] OR sonograp*[tw] OR us [sh]) OR (Magnetic Resonance Imaging [mh] OR MR imag*[tw] OR magnetic resonance imag* [tw] OR MRI [tw])) AND (Rotator Cuff [mh] OR rotator cuff* [tw] OR musculotendinous cuff* [tw] OR subscapularis [tw] OR supraspinatus [tw] OR infraspinatus OR teres minor [tw]) AND (Rupture [mh:noexp] OR tear* [tw] OR torn [tw] OR thickness [tw] OR lesion* [tw] OR ruptur* [tw] OR injur* [tw]))

From Lenza, M., Buchbinder, R., Takwoingi, Y., Johnston, R. V., Hanchard, N. C., & Faloppa, F. (2013). Magnetic resonance imaging, magnetic resonance arthrography and ultrasonography for assessing rotator cuff tears in people with shoulder pain for whom surgery is being considered. The Cochrane Library.

The Boolean model: summary

- Documents either match or don't match
 - ◇ Expert knowledge needed to create high-precision queries → OK for expert users
 - ◇ Often used by bibliographic search engines (library)
- Not good for the majority of users
 - ◇ Most users not familiar with writing Boolean queries → not natural
 - ◇ Most users don't want to wade through 1000s of unranked results → unless very specific search in small collections
 - ◇ This is particularly true of web search → very large set of docs

The Vector Space model

- Documents are also represented as “bags of words”:
 - ◇ “John is quicker than Mary” = “Mary is quicker than John”
- Documents are points in high-dimensional vector space
 - ◇ each term in index is a dimension → sparse vectors
 - ◇ values are frequencies of terms in documents, or variants of frequency
- Queries are also represented as vectors (for terms that exist in index)
- Approach
 - ◇ Document–query similarity is the basis for relevance (ranking)
 - ◇ Select document(s) with highest document–query similarity
 - ◇ Present documents to user in descending order by relevance to query
 - ◇ With ranking, the number of returned documents is less relevant → users start at the top of the ranking and stop when satisfied

The Vector Space Model (contd)

Approach:

- Think of document collection as an $m \times n$ document-term matrix M
 - ◊ m is the number of documents in the collection, and
 - ◊ n is the number of distinct terms that occur across all documents
- Each row of M is a document vector, with each vector element $M_{i,j}$ a term weight (e.g. the count) of term j in document i
- For example:

	Term ₁	Term ₂	Term ₃	...	Term _n
Doc ₁	9	0	1	...	0
Doc ₂	0	1	0	...	10
Doc ₃	0	1	0	...	2
...	...	...	...	...	...
Doc _m	4	7	0	...	5

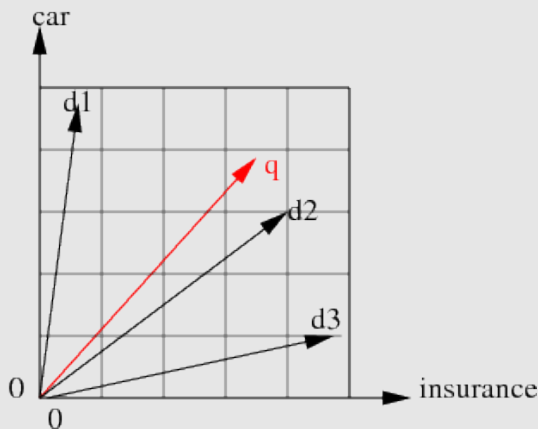
Q	0	1	0	...	1
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- To rank documents according to their **similarity** to the query
 - ◊ compare vector of **query** against vector of each **document**

The Vector Space model (contd)

Simplified example in 2 dimensions:

Query: car insurance



How to measure similarity between vectors?

- Each document and the query are represented as a vector of n values:

$$\vec{d}^i = (d_1^i, d_2^i, \dots, d_n^i), \quad \vec{q} = (q_1, q_2, \dots, q_n)$$

- Many metrics of similarity between 2 vectors, e.g.: [Euclidean](#)

$$\sqrt{\sum_{k=1}^n (q_k - d_k)^2}$$

- E.g. (from previous slide): Distance between:

$$Doc_1 \text{ and } Q = \sqrt{(9-0)^2 + (0-1)^2 + (1-0)^2 + (0-1)^2} = \sqrt{84} = 9.15$$

$$Doc_2 \text{ and } Q = \sqrt{(0-0)^2 + (1-1)^2 + (0-0)^2 + (10-1)^2} = \sqrt{81} = 9$$

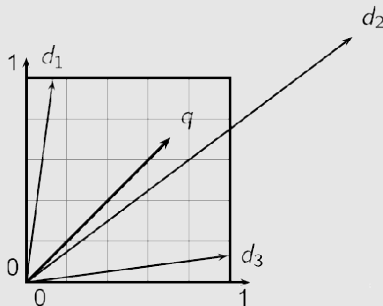
$$Doc_3 \text{ and } Q = \sqrt{(0-0)^2 + (1-1)^2 + (0-0)^2 + (2-1)^2} = \sqrt{1} = 1$$

Doc 3 is the closest (shortest distance)

How to measure similarity between vectors? (contd)

Is it a good idea?

- distance is large for vectors of different lengths, even if by only one term (e.g. Doc_2 and Q)
- means frequency of terms given *too much impact*



- NB: this figure is **not** meant to be an accurate representation of the example

How to measure similarity between vectors? (contd)

- Better **similarity** metric, used in *vector-space* model: **cosine** of the **angle** between two vectors $\vec{x}$ and $\vec{y}$:

$$\cos(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}||\vec{y}|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}}$$

- It can be interpreted as the **normalised correlation coefficient**:
 - i.e. it computes how well the x_i and y_i correlate, and then divides by the length of the vectors, to scale for their magnitude
 - ◇ The vector $\vec{x}$ is normalised by dividing its components by its length:

$$|\vec{x}| = \sqrt{\sum_{i=1}^n x_i^2}$$

How to measure similarity between vectors? (contd)

- The cosine value ranges from:
 - ◇ 1, for vectors pointing in the same direction, to
 - ◇ 0, for orthogonal vectors, to
 - ◇ -1, for vectors pointing in opposite directions
- Specialising the equation to comparing a query q and document d :

$$\text{sim}(\vec{q}, \vec{d}) = \cos(\vec{q}, \vec{d}) = \frac{\sum_{i=1}^n q_i d_i}{\sqrt{\sum_{i=1}^n q_i^2} \sqrt{\sum_{i=1}^n d_i^2}}$$

- i.e. computes how well occurrences of each term i correlate in query and document, then scales for the magnitude of the overall vectors

The Probabilistic Retrieval Model

- Like the vector space model (VSM), the Probabilistic Retrieval Model (PRM) is a ranked retrieval model
 - ◊ i.e. the match between query and document a matter of degree
- However
 - ◊ the VSM measures the relatedness of query and document in terms of similarity of query and document vectors,
 - ◊ the PRM measures it in terms of the probability that a document is relevant to a given query
- The PRM generally viewed as being theoretically better founded than the VSM, as it is grounded in probability theory
 - ◊ However, simplifying assumptions needed to estimate the required probabilities result in the PRM being in practice v. similar to the VSM
 - ◊ Mention only because it led the popular BM25 document-query ranking function (see next lecture)

Summary

- For purposes of retrieval documents are generally treated as **bags-of-words**
- Three well-known retrieval models are:
 - ◇ the **Boolean Model**
 - each document in a collection is either relevant to a given query or not relevant to that query
 - queries are formulated using boolean operators (AND, OR, NOT) between terms
 - ◇ the **Vector Space Model**
 - each document in a collection has some degree of relevance to query
 - documents and queries are both represented as vectors of the terms occurring within them
 - degree of similarity computed as cosine of angle between document and query vectors
 - ◇ the **Probabilistic Model**
 - ranked retrieval model like the vector space model
 - relatedness of document to query measured as the probability that a document is relevant to a given query